Quiz 11: Blood Spatter & Glass

Review of Session 11. ~15 minutes. Calculator with arcsin needed.

Section A: Spatter shape & velocity

1. A drop hits at an angle and makes an elongated stain with a pointed tail. Which way was the drop traveling?
2. Match the velocity category to the cause:
    - large drops (>4 mm), drips and pools
    - medium drops (1–4 mm)
    - fine mist (<1 mm)
3. What is a "void" in a spatter pattern, and what does it tell you?

Section B: Angle of impact

1. Write the formula for the angle of impact.
2. A stain is 4 mm wide and 8 mm long. What is the angle of impact?
3. A stain is 5 mm wide and 5 mm long. What is the angle of impact?
4. What is the "area of convergence," and how do you find it?
5. What extra information do you need to find the area of *origin*?

Section C: Glass

1. Which cracks form first — radial or concentric?
2. Radial cracks form on which side relative to the force?
3. State the 3R rule.
4. A bullet hole in glass is narrow on one side and wide on the other. Which way was the bullet traveling?
5. Two impact holes: cracks from hole B stop where they meet cracks from hole A. Which impact happened first?
6. A pane with a clean hole and few cracks vs. a shattered pane with many long radial cracks — which was struck by a high-velocity projectile?